

Um Novo Modelo Dinâmico Bidimensional para Modelagem de Imagens SAR

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Resumo

Na área de sensoriamento remoto, é comum o uso de sensores ativos, como os sistemas de radar de abertura sintética (SAR). O conhecimento de propriedades estatísticas e a modelagem de imagens SAR possuem papel de destaque no monitoramento do uso do solo, pois esse tipo de imagem apresenta alta qualidade, independentemente de características climáticas, dia ou noite. Neste trabalho, é introduzido um novo modelo bidimensional (2D) autorregressivo (AR) capaz de modelar a estrutura de correlação espacial dos pixels de imagens SAR no formato de amplitude. Como a amplitude de dados SAR possui valores contínuos positivos, é pressuposto que, condicionalmente a alguma vizinhança espacial, esses valores advêm de realizações de variáveis aleatórias com distribuição Maxwell. A estimação dos parâmetros do modelo Maxwell bidimensional autorregressivo (2D-MxAR), reparametrizado em termos da média, é realizada utilizando o método da máxima verossimilhança condicional (MVC). Expressões matriciais simples para o vetor escore e a matriz de informação de Fisher são obtidas. Inferências intervalares e testes de hipóteses são discutidos com base na distribuição assintótica dos estimadores de MVC. A avaliação do desempenho dos estimadores é conduzida por meio de simulações de Monte Carlo, que usa observações pseudoaleatórias geradas pelo método da inversão.

1 Introduction

2 The proposed model

The bidimensional autoregressive framework has already been introduced in the literature (see Palm et al. (2022) and Zielinski et al. (2010)), where the variable Y is assumed to be spatially autocorrelated in two dimensions.

Let $Y[n, m]$, with $n = 1, 2, \dots, N$ and $m = 1, 2, \dots, M$, denote a random variable representing the pixel intensity at location (n, m) in an $N \times M$ image. Define the strongly causal region at $[n, m]$ as

$$S[n, m] = \{(k, l) \in \mathbb{Z}^2 : 1 \leq k \leq n, 1 \leq l \leq m\} - \{(n, m)\}, \quad (1)$$

following the definition in ?. Conditionally on the information set in the neighborhood $S[n, m]$, each $Y[n, m]$ is assumed to follow the Maxwell distribution.

Adopting the reparameterization in terms of the mean proposed by Lemonte (2024), the conditional density of $Y[n, m]$, given $S[n, m]$, is expressed as

$$f(y_{n,m} | S[n, m]) = \frac{32y_{n,m}^2}{\pi^2 \mu_{n,m}^3} \exp\left(-\frac{4y_{n,m}^2}{\pi \mu_{n,m}^2}\right), \quad (2)$$

and the cumulative distribution function is given by

$$F(y_{n,m} | S[n, m]) = \frac{2\gamma(3/2, 4y_{n,m}^2/(\pi \mu_{n,m}^2))}{\sqrt{\pi}},$$

where $\mu_{n,m} > 0$ and

$$\gamma(a, x) = \int_0^x t^{a-1} \exp(-t) dt$$

is the lower incomplete gamma function. The conditional variance of $Y_{n,m}$, in terms of $\mu_{n,m}$, is given by

$$\text{Var}(Y_{n,m} \mid S[n, m]) = 0.178 \mu_{n,m}^2,$$

and the conditional mean is

$$\mathbb{E}(Y_{n,m} \mid S[n, m]) = \mu_{n,m}.$$

2.1 The 2D-MxARMA model

The dynamic structure of the bidimensional Maxwell ARMA model, denoted as 2D-MxARMA(p, q), is defined through a monotonic link function. We employ a linear predictor represented as

$$\eta_{n,m} = g(\mu_{n,m}),$$

where $g(\cdot)$ is a monotonic and twice differentiable link function, such that $g : \mathbb{R}^+ \rightarrow \mathbb{R}$. Thus, the model can be written as

$$g(\mu_{n,m}) = \alpha + \sum_{i=1}^p \sum_{j=1}^p \phi_{i,j} g(y_{n-i, m-j}) + \sum_{k=1}^q \sum_{l=1}^q \theta_{k,l} r_{n-k, m-l}, \quad (3)$$

where β is the intercept, p and q are the autoregressive and moving average orders, $\phi_{i,j}$ and $\theta_{k,l}$ are the corresponding coefficients, and $r_{n,m}$ denotes the backward error term.

Under these conditions, the proposed 2D-MxARMA model inherits the desirable properties of autoregressive moving average processes, such as identifiability, stationarity, ensuring the consistency and asymptotic normality of the conditional maximum likelihood estimators (see Brockwell and Davis (1991); ?). These results provide the theoretical foundation for the inference procedures discussed in the next section.

3 Conditional likelihood inference

The inference of the parameters of the 2D-MxARMA model is carried out by means of the conditional maximum likelihood estimation method (Casella and Berger, 2002). Let

$$\boldsymbol{\gamma} = (\alpha, \phi^\top, \theta^\top)^\top$$

be the parameter vector, where $\phi = (\phi_{0,1}, \phi_{0,2}, \dots, \phi_{p,p})^\top$ and $\theta = (\theta_{0,1}, \theta_{0,2}, \dots, \theta_{q,q})^\top$, with dimensions $(p+1)^2 - 1$ and $(q+1)^2 - 1$, respectively.

The conditional log-likelihood function, given $w = \max(p, q)$ preliminary observations, is defined as

$$\ell(\boldsymbol{\gamma}) = \sum_{n=w+1}^N \sum_{m=w+1}^M \log f_Y(y_{n,m} \mid S[n, m]) = \sum_{n=w+1}^N \sum_{m=w+1}^M \ell_{n,m}(\mu_{n,m}), \quad (4)$$

where

$$\ell_{n,m}(\mu_{n,m}) = \log(32) - 2 \log(\pi) - 3 \log(\mu_{n,m}) + \log(y_{n,m}^2) - \frac{4y_{n,m}^2}{\pi \mu_{n,m}^2}. \quad (5)$$

The conditional maximum likelihood estimators (CMLEs) are obtained by solving the score equations

$$\mathbf{U}(\boldsymbol{\gamma}) = \frac{\partial \ell}{\partial \boldsymbol{\gamma}^\top} = \left(\frac{\partial \ell}{\partial \alpha}, \frac{\partial \ell}{\partial \phi^\top}, \frac{\partial \ell}{\partial \theta^\top} \right)^\top = \mathbf{0}, \quad (6)$$

where $\mathbf{U}(\boldsymbol{\gamma})$ is a vector of dimension $(p+1)^2 + (q+1)^2 - 1$.

From the chain rule decomposition, we obtain

$$\frac{\partial \ell}{\partial \boldsymbol{\gamma}} = \sum_{n=w+1}^N \sum_{m=w+1}^M \frac{\partial \ell_{n,m}(\mu_{n,m})}{\partial \mu_{n,m}} \cdot \frac{d\mu_{n,m}}{d\eta_{n,m}} \cdot \frac{\partial \eta_{n,m}}{\partial \boldsymbol{\gamma}}. \quad (7)$$

The required derivatives are

$$\frac{\partial \ell_{n,m}(\mu_{n,m})}{\partial \mu_{n,m}} = \frac{8y_{n,m}^2}{\pi \mu_{n,m}^3} - \frac{3}{\mu_{n,m}}, \quad \frac{d\mu_{n,m}}{d\eta_{n,m}} = \frac{1}{g'(\mu_{n,m})} = \mu_{n,m}, \quad (8)$$

since the link function $g(\cdot)$ is logarithmic.

The derivatives of $\eta_{n,m}$ with respect to each parameter are given by

$$\frac{\partial \eta_{n,m}}{\partial \alpha} = 1 - \sum_{s=0}^q \sum_{t=0}^q \theta_{(s,t)} \frac{\partial \eta_{n-s,m-t}}{\partial \alpha}, \quad (9)$$

$$\frac{\partial \eta_{n,m}}{\partial \phi_{(i,j)}} = g(y_{n-i,m-j}) - \sum_{s=0}^q \sum_{t=0}^q \theta_{(s,t)} \frac{\partial \eta_{n-s,m-t}}{\partial \phi_{(i,j)}}, \quad (10)$$

$$\frac{\partial \eta_{n,m}}{\partial \theta_{(k,l)}} = g(y_{n-k,m-l}) - g(\mu_{n-k,m-l}) - \sum_{s=0}^q \sum_{t=0}^q \theta_{(s,t)} \frac{\partial \eta_{n-s,m-t}}{\partial \theta_{(k,l)}}, \quad (11)$$

for $(i,j) \in \{0,1,\dots,p\}^2 \setminus \{(0,0)\}$ and $(k,l) \in \{0,1,\dots,q\}^2 \setminus \{(0,0)\}$.

Since the likelihood equations do not admit closed-form solutions, numerical methods are required to obtain the CMLEs. In this work, we employ the quasi-Newton **Broyden-Fletcher-Goldfarb-Shanno (BFGS)** method (Wright et al., 1999), which is widely recognized for its efficiency in nonlinear optimization problems.

The initial values of the parameters are derived from linear regression estimates applied to the transformed values $g(y)$ for α and ϕ_p , while the moving average parameters θ_q are set to zero by default, as suggested in Bayer et al. (2020).

Moreover, the analytical derivatives presented in (8) are used to construct the gradient, which improves both stability and convergence speed of the algorithm. Thus, the CMLEs are obtained as the solution of the optimization problem

$$\hat{\gamma} = \arg \max_{\gamma \in \Gamma} \ell(\gamma).$$

3.1 Conditional Fisher information matrix

The Fisher information matrix is a symmetric and positive-definite matrix that plays a central role in statistical inference. It can be used to derive the asymptotic covariance matrix of the conditional maximum likelihood estimators (CMLEs), which in turn is essential for hypothesis testing and the construction of test statistics such as the Wald test.

Formally, the Fisher information matrix is defined as the expected value of the negative Hessian of the log-likelihood function, or equivalently as the expected outer product of the score vector:

$$\mathbf{K}(\gamma) = \mathbb{E} \left[\frac{\partial \ell(\gamma)}{\partial \gamma} \frac{\partial \ell(\gamma)}{\partial \gamma^\top} \middle| S \right] = \left\{ \mathbb{E} \left[\frac{\partial \ell(\gamma)}{\partial \gamma_i} \frac{\partial \ell(\gamma)}{\partial \gamma_j} \middle| S \right] \right\} = \left\{ -\mathbb{E} \left[\frac{\partial^2 \ell(\gamma)}{\partial \gamma_i \partial \gamma_j} \middle| S \right] \right\}, \quad (12)$$

for $i, j = 1, \dots, k + (p+1)^2 + (q+1)^2$.

In our case, the conditional expectation of the second derivative of the log-likelihood is given by

$$\mathbb{E} \left(\frac{\partial^2 \ell(\gamma)}{\partial \gamma_i \partial \gamma_j} \middle| S[n, m] \right) = \sum_{n=1}^N \sum_{m=1}^M \mathbb{E} \left(\frac{\partial^2 \ell_{n,m}(\mu_{n,m})}{\partial \mu_{n,m}^2} \right) \left(\frac{d\mu_{n,m}}{d\eta_{n,m}} \right)^2 \frac{\partial \eta_{n,m}}{\partial \gamma_j} \frac{\partial \eta_{n,m}}{\partial \gamma_i}. \quad (13)$$

Analogously to the first derivative in (8), the second derivative of $\ell_{n,m}(\mu_{n,m})$ with respect to $\mu_{n,m}$ can be evaluated as

$$\begin{aligned} \mathbb{E} \left(\frac{\partial^2 \ell_{n,m}(\mu_{n,m})}{\partial \mu_{n,m}^2} \middle| \mathcal{S}_{n,m} \right) &= \frac{24(0.178\mu_{n,m}^2 + \mu_{n,m}^2)}{\mu_{n,m}^4 \pi} + \frac{3}{\mu_{n,m}^2} \\ &= -\frac{6}{\mu_{n,m}^2}, \end{aligned}$$

and, since $g(\cdot)$ is logarithmic,

$$\left(\frac{d\mu_{n,m}}{d\eta_{n,m}} \right)^2 = \mu_{n,m}^2.$$

Therefore,

$$\mathbb{E} \left(\frac{\partial^2 \ell(\gamma)}{\partial \gamma_i \partial \gamma_j} \middle| S[n, m] \right) = \sum_{n=1}^N \sum_{m=1}^M -\frac{6}{\mu_{n,m}^2} \mu_{n,m}^2 \frac{\partial \eta_{n,m}}{\partial \gamma_j} \frac{\partial \eta_{n,m}}{\partial \gamma_i}. \quad (14)$$

The conditional Fisher information matrix for the parameter vector $\boldsymbol{\gamma}$ can then be expressed in block form as

$$\mathbf{K}(\boldsymbol{\gamma}) = \begin{pmatrix} K(\alpha, \alpha) & K(\alpha, \phi) & K(\alpha, \theta) \\ K(\phi, \alpha) & K(\phi, \phi) & K(\phi, \theta) \\ K(\theta, \alpha) & K(\theta, \phi) & K(\theta, \theta) \end{pmatrix}, \quad (15)$$

where

$$\begin{aligned} \mathbf{K}(\alpha, \alpha) &= \mathbf{a}^\top \mathbf{W} \mathbf{a}, & \mathbf{K}(\alpha, \phi) &= \mathbf{a}^\top \mathbf{W} \mathbf{P}, & \mathbf{K}(\alpha, \theta) &= \mathbf{a}^\top \mathbf{W} \mathbf{R}, \\ \mathbf{K}(\phi, \alpha) &= \mathbf{P}^\top \mathbf{W} \mathbf{a}, & \mathbf{K}(\phi, \phi) &= \mathbf{P}^\top \mathbf{W} \mathbf{P}, & \mathbf{K}(\phi, \theta) &= \mathbf{P}^\top \mathbf{W} \mathbf{R}, \\ \mathbf{K}(\theta, \alpha) &= \mathbf{R}^\top \mathbf{W} \mathbf{a}, & \mathbf{K}(\theta, \phi) &= \mathbf{R}^\top \mathbf{W} \mathbf{P}, & \mathbf{K}(\theta, \theta) &= \mathbf{R}^\top \mathbf{W} \mathbf{R}. \end{aligned}$$

Here, the matrices $\mathbf{P}$ and $\mathbf{R}$ have dimensions $(NM - w) \times ((p+1)^2 - 1)$ and $(NM - w) \times ((q+1)^2 - 1)$, respectively, with elements

$$\mathbf{P}[i, j] = \frac{\partial \eta_{i+w, j+w}}{\partial \phi_{i, j}}, \quad \mathbf{R}[i, j] = \frac{\partial \eta_{i+w, j+w}}{\partial \theta_{i, j}}.$$

The vector $\mathbf{a}$ is defined as

$$\mathbf{a} = \left(\frac{\partial \eta_{w+1, w+1}}{\partial \alpha}, \frac{\partial \eta_{w+2, w+2}}{\partial \alpha}, \dots, \frac{\partial \eta_{N, M}}{\partial \alpha} \right)^\top.$$

Finally, the weight matrix $\mathbf{W}$ can be simplified as a diagonal matrix:

$$\mathbf{W} = \text{diag} \left\{ \frac{6\mu_{1,1}^2}{\mu_{1,1}^2}, \frac{6\mu_{2,1}^2}{\mu_{2,1}^2}, \dots, \frac{6\mu_{N,M}^2}{\mu_{N,M}^2} \right\} = 6\mathbf{I}_{NM-w}, \quad (16)$$

where $\mathbf{I}_{NM-w}$ is the identity matrix of dimension $(NM - w)$.

3.2 Confidence Intervals and Hypothesis Testing

Based on the Fisher Information, it is possible to construct confidence intervals and perform hypothesis testing for the CMLEs, provided that the standard regularity conditions hold Lehmann and Casella, 1998. From the asymptotic theory of conditional maximum likelihood estimators, we obtain the following convergence result:

$$(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \xrightarrow[N \cdot M \rightarrow \infty]{d} \mathcal{N}(\mathbf{0}, \mathbf{K}^{-1}(\boldsymbol{\gamma})),$$

where $\xrightarrow{d}$ denotes convergence in distribution, and $\mathbf{K}^{-1}(\boldsymbol{\gamma})$ represents the asymptotic covariance matrix of the CMLEs, derived from the inverse of the Fisher Information matrix. This result ensures that, asymptotically, the CMLEs are unbiased, consistent, and asymptotically normal.

The significance of the estimated parameters in the two-dimensional MxARMA model can be evaluated through standard Wald-type hypothesis tests. Specifically, we test

$$\begin{cases} \mathcal{H}_0 : \boldsymbol{\gamma}^* = \mathbf{0}, \\ \mathcal{H}_1 : \boldsymbol{\gamma}^* \neq \mathbf{0}, \end{cases}$$

where $\boldsymbol{\gamma}^* = (\alpha, \phi^\top, \theta^\top)^\top$. Under the null hypothesis, the Wald statistic

$$T_w = (\hat{\boldsymbol{\gamma}}^*)^\top \left[\widehat{\text{Var}}(\hat{\boldsymbol{\gamma}}^*) \right]^{-1} \hat{\boldsymbol{\gamma}}^*$$

follows, asymptotically, a χ^2 -distribution with degrees of freedom equal to the number of tested parameters. This allows for both individual and joint significance testing of the coefficients.

Additionally, a confidence interval for the i th component of $\boldsymbol{\gamma}$, $i = 1, 2, \dots, (p+1)^2 + (q+1)^2 - 1$, with confidence level approximately $100(1 - \alpha)\%$, can be derived from the CMLE asymptotic distribution as

$$[\hat{\gamma}_i - z_{1-\alpha/2} \widehat{\text{se}}(\hat{\gamma}_i), \hat{\gamma}_i + z_{1-\alpha/2} \widehat{\text{se}}(\hat{\gamma}_i)],$$

where

$$\widehat{\text{se}}(\hat{\gamma}_i) = \sqrt{[\widehat{\mathbf{K}}^{-1}(\hat{\boldsymbol{\gamma}})]_{ii}}$$

is the standard error (SE) of $\hat{\gamma}_i$, $[\widehat{\mathbf{K}}^{-1}(\hat{\boldsymbol{\gamma}})]_{ii}$ is the i th diagonal element of the estimated asymptotic covariance matrix $\widehat{\mathbf{K}}^{-1}(\hat{\boldsymbol{\gamma}})$, and $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard normal distribution.

{Arthur: conferir o teste de wald}

4 Diagnostic analysis and Anomaly Detection

In this section, we discuss the diagnostic tools employed to evaluate the adequacy of the proposed 2D-MxARMA model (3). Such diagnostics are essential not only to assess model fit but also to enable anomaly detection in spatio-temporal data. A common approach is based on the analysis of *quantile residuals* (Dunn and Smyth, 1996), which are defined as

$$e_{n,m} = \Phi^{-1}(F_Y(y_{n,m}; \hat{\mu}_{n,m} | S_{n,m})), \quad (17)$$

where Φ^{-1} denotes the quantile function of the standard normal distribution and $F_Y(\cdot; \hat{\mu}_{n,m} | S_{n,m})$ is the fitted conditional cumulative distribution function of $Y_{n,m}$. By construction, the residuals $e_{n,m}$ are approximately distributed as $\mathcal{N}(0, 1)$ under correct model specification. Therefore, diagnostic evaluation involves verifying whether the residuals are uncorrelated and conform to the standard Gaussian distribution.

The predictive structure of the fitted model is given by

$$\hat{\mu}_{n,m} = g^{-1} \left(\hat{\alpha} + \sum_{i=1}^p \sum_{j=1}^p \hat{\phi}_{i,j} g(y_{n-i,m-j}) + \sum_{k=1}^q \sum_{l=1}^q \hat{\theta}_{k,l} r_{n-k,m-l} \right), \quad (18)$$

with $n = w + 1, w + 2, \dots, N$ and $m = w + 1, w + 2, \dots, M$. Hence, the boundary values are excluded from the estimation, resulting in a fitted matrix of size $(N - w) \times (M - w)$.

To evaluate model adequacy, two key aspects of the residuals must be verified:

- **Normality:** since the quantile residuals should follow a standard Gaussian distribution, graphical methods such as histograms and QQ-plots, as well as formal tests (e.g., Shapiro–Wilk), can be used to check whether the assumption holds.
- **Independence:** the absence of remaining spatio-temporal autocorrelation can be assessed through residual correlograms or generalized portmanteau-type tests, ensuring that the model captures the dependence structure adequately.

These checks provide evidence of whether the fitted model is consistent with the theoretical assumptions underlying the 2D-MxARMA framework.

In particular, the use of quantile residuals makes it possible to perform standard Gaussian-based diagnostic analyses in a generalized distributional framework, thus providing a coherent tool for both model validation and anomaly detection in two-dimensional time series.

For testing independence, a two-dimensional version of the Ljung–Box statistic can be defined as

$$Q(h_x, h_y) = (N - w)(M - w) \sum_{i=1}^{h_x} \sum_{j=1}^{h_y} \frac{\hat{\rho}^2(i, j)}{N \cdot M}, \quad (19)$$

where $\hat{\rho}(i, j)$ denotes the sample autocorrelation of the residuals at spatial lag (i, j) and (h_x, h_y) are the maximum lags considered in each direction. Under the null hypothesis of independence, $Q(h_x, h_y)$ is asymptotically distributed as a chi-square random variable with $(h_x \cdot h_y)$ degrees of freedom (?).

These checks provide evidence of whether the fitted model is consistent with the theoretical assumptions underlying the 2D-MxARMA framework.

5 Simulation Results

This section presents two complementary Monte Carlo experiments designed to evaluate the behavior of the proposed 2D-MxARMA model under controlled conditions. The first study investigates the model's *detection power*, quantifying its ability to identify true targets embedded in simulated images. The second study examines the *numerical performance of the CMLEs*, focusing on bias and mean squared error as the sample size increases. All computations were carried out in the R programming language R Core Team (2024), and the corresponding code is publicly available at: <https://github.com/arthurhintz/MxARMA>.

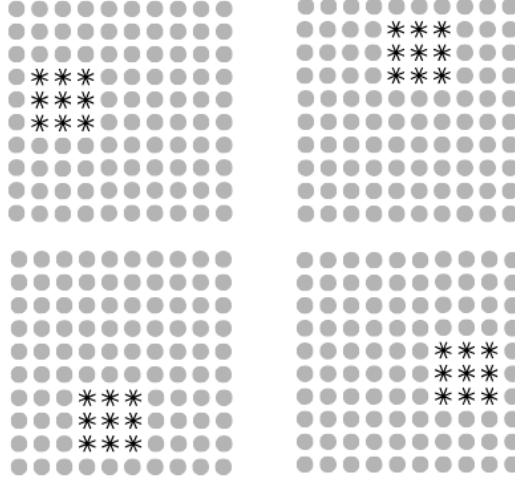


Figure 1: Spatial arrangement of the target in the simulation.

5.1 Detection Power

This simulation assesses whether the proposed methodology can effectively identify targets artificially introduced into synthetic images. A 10×10 matrix was generated from a 2D-MxARMA(1,1) model, where the coefficients ϕ and θ were drawn independently from the interval $(-0.3, 0.3)$, and the α parameter was sampled from $U(0, 0.6)$. Nine target locations were fixed in a 3×3 pattern, and their intensities were sampled from $U(1, 40)$. The model was then fitted to each of the four rotations of the image. A target was declared detected at a given position if *at least one rotation* produced a quantile residual with absolute value greater than 3.

Figure 1 displays the pattern of target locations used in the simulation.

The complete simulation procedure is summarized as follows:

1. Define the matrix dimension n and the number of Monte Carlo replications n_{rep} .
2. Construct the set of fixed target positions P for the four rotations.
3. For each replication $i = 1, \dots, n_{\text{rep}}$:
 - (a) Generate target intensities $w_j \sim U(1, 40)$.
 - (b) Sample $\alpha \sim U(0, 0.6)$.
 - (c) Sample the ϕ and θ parameters by selecting three values uniformly from a predefined set x .
 - (d) Simulate a field $\mathbf{Y}$ from the 2D-MxARMA model with parameters (α, ϕ, θ) .
 - (e) Insert the target intensities w_j in the positions $P_{1, \cdot}$ of $\mathbf{Y}$.
 - (f) For each rotation $j = 1, \dots, 4$:
 - i. Fit the 2D-MxARMA model to the rotated image.
 - ii. Compute the residuals $\mathbf{e}$.
 - iii. Binarize the residuals as $e_{ij}^{(b)} = 1$ if $|e_{ij}| > 3$, and 0 otherwise.
 - iv. Record $R_{j,k} = e_{P_{j,k}}^{(b)}$ for each target position k .
 - (g) Compute the number of detections at each target location,

$$d_k = \sum_{j=1}^4 R_{j,k}.$$

- (h) Mark target k as detected if $d_k \geq 1$.

(i) Compute the type-II error for replication i :

$$\text{Error}_i = 1 - \frac{\text{number of detected targets}}{\text{total number of targets}}.$$

The detection power is then given by $1 - \text{Error}_i$, representing the probability of correctly identifying a true target.

Os resultados indicaram ...

5.2 Estimação numérica

This subsection presents a Monte Carlo study designed to evaluate the finite-sample performance of the conditional maximum likelihood estimators (CMLEs) of the 2D-MxARMA model. A total of 5000 replications were generated for matrix sizes $N = K \in \{10, 30, 50, 100\}$, using the logarithmic link function.

The true parameter values were chosen to match those obtained in the real-data application $\alpha = -1.2$, $\phi_{0,0} = 0.030$, $\phi_{0,1} = 0.250$, $\phi_{1,0} = 0.350$, $\theta_{0,0} = -0.100$, $\theta_{0,1} = -0.008$, and $\theta_{1,0} = -0.060$. To evaluate estimator performance, we computed the root mean squared error (RMSE) and mean squared error (MSE).

As expected for consistent estimators, both RMSE and MSE decrease as the matrix size increases. Table 1 summarizes the results. The mean CMLEs converge toward the true parameter values as N grows, and the reduction in MSE across all parameters provides strong numerical evidence of consistency.

Tabela 1: Monte Carlo Simulation Results for the 2D-MxARMAREG(1,1)

Measures	N = M	Mean	RMSE	MSE
$\alpha = -1.200$	10	-1.4682	1.5179	0.2893
	30	-1.3010	1.2975	0.0383
	50	-1.2398	1.2340	0.0165
	100	-1.2243	1.2179	0.0065
$\phi_{0,0} = 0.030$	10	0.0346	0.5206	0.0313
	30	0.0307	0.4986	0.0046
	50	0.0222	0.4894	0.0022
	100	0.0249	0.4918	0.0006
$\phi_{0,1} = 0.250$	10	0.2047	0.5971	0.0365
	30	0.2335	0.5855	0.0057
	50	0.2499	0.5915	0.0026
	100	0.2503	0.5907	0.0008
$\phi_{1,0} = 0.350$	10	0.3086	0.6489	0.0322
	30	0.3346	0.6488	0.0044
	50	0.3458	0.6535	0.0019
	100	0.3474	0.6562	0.0006
$\theta_{0,0} = -0.100$	10	-0.0925	0.5010	0.0382
	30	-0.0911	0.4804	0.0038
	50	-0.0906	0.4752	0.0015
	100	-0.0936	0.4717	0.0004
$\theta_{0,1} = -0.008$	10	0.0230	0.5317	0.0466
	30	0.0095	0.4821	0.0061
	50	-0.0078	0.4895	0.0028
	100	-0.0081	0.4827	0.0009
$\theta_{1,0} = -0.060$	10	-0.0333	0.5197	0.0444
	30	-0.0431	0.4804	0.0052
	50	-0.0556	0.4799	0.0022
	100	-0.0575	0.4766	0.0007

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